# Fine-tuning NVIDIA GenMol

## A Proof of Concept for Small-Molecule Design on Delaney / ESOL

Discrete-diffusion generation over SAFE sequences · One A100 GPU via Slurm

Objective: generate valid, novel, drug-like molecules aligned to the target chemical domain

**Pipeline: prepare → fine-tune → generate / evaluate**

PROJECT OVERVIEW

# Objective & Deliverables

## Objective

- Fine-tune the pretrained NVIDIA GenMol model on a focused small-molecule dataset
- Evaluate whether it generates valid, novel, drug-like molecules aligned to the domain
- Must run on one A100 GPU managed through Slurm
- **Emphasis is on approach, not accuracy alone**

## Deliverables

- Source code + environment setup instructions
- Slurm submission scripts
- Data-preparation script + dataset description
- Presentation: dataset & preprocessing, fine-tuning config, base vs fine-tuned comparison, five example molecules, limitations, next steps

APPROACH

# Pipeline & Design Decisions

### 1 · Prepare

Canonicalize, validity-filter,
dedupe, drug-like bounds,
Bemis–Murcko scaffold split,
SMILES ⟶ SAFE

### 2 · Fine-tune

Load pretrained GenMol + EMA,
lr 1e-4, up to 25 epochs,
validate every epoch,
select min val-loss best.ckpt

### 3 · Generate / Evaluate

Base vs fine-tuned,
shared seeded length prior,
validity · novelty · QED · SA ·
diversity · similarity · FCD

## Why these choices matter

- Frozen, scaffold-disjoint split — validation scaffolds never appear in training; written once and reused.
- Real validation — val.safe scored every epoch, averaging 4 diffusion-time samples to cut Monte-Carlo noise.
- Controlled sampling — base and fine-tuned draw from the same seeded SAFE length prior, isolating token-level changes.

# Beyond the Assignment — What This PoC Adds

The brief asked for a fine-tuning run, Slurm scripts, data prep, and a presentation. This PoC also adds:

- **Scaffold-disjoint validation**
  - Bemis–Murcko split keeps val scaffolds out of training; --split random fallback
- **Reliable checkpoint selection**
  - Val every epoch, multi diffusion-time averaging, lowest-val-loss best.ckpt
- **Upstream loader-bug workaround**
  - Scalar-indexed dataset instead of broken UserDataset; vendored GenMol untouched
- **Controlled generation lengths**
  - Identical seeded SAFE-length draws isolate token-level changes from size
- **Expanded evaluation**
  - FCD, nearest-val similarity, diversity, QED/MW/logP overlays, 5-molecule grid
- **Small-data regularization**
  - --freeze-layers adapts only upper layers + MLM head, EMA-compatible
- **Hyperparameter sweep**
  - Slurm array over freeze × lr; aggregate_sweep.py gates + ranks survivors
- **Offline experiment tracking**
  - Per-run summary, loss history, checkpoint metadata, central run registry
- **Held-out generation metrics**
  - Scaffold-held-out split drives val loss, selection, similarity, and FCD
- **Reproducibility controls**
  - All stages seeded; deterministic Lightning where CUDA kernels allow
- **End-to-end V1 / V2 support**
  - Checkpoint paths, SAFE mode, and sampling defaults switch together
- **Smoke-test path**
  - SMOKE=1 runs a short end-to-end check before a full A100 job

# Dataset & Preprocessing

- Source: DeepChem Delaney (ESOL) aqueous-solubility dataset — 1,128 molecules
- Uses only the SMILES column; solubility labels are not trained on or predicted
- RDKit canonicalize ⟶ validity filter ⟶ dedupe ⟶ drug-like bounds (5–60 heavy atoms, MW ≤ 750) ⟶ SAFE encoding
- Deterministic 90/10 Bemis–Murcko scaffold split

## Domain descriptors

| Descriptor | Mean | Range |
|---|---|---|
| Molecular weight | 209 Da | 68–589 |
| logP | 2.53 | -7.6–10.4 |
| QED | 0.56 | 0.15–0.93 |
| Rings | 1.45 | 0–7 |
| Heavy atoms | 13.7 | 5–42 |

| Stage | Molecules |
|---|---|
| Raw records | 1,128 |
| Duplicates removed | 11 |
| Out-of-range removed | 61 |
| Final dataset | 1,056 |
| Train / validation | 951 / 105 |

### Nature of the domain

A broad, compact, low-MW, lead-like distribution — not a target-specific chemotype.

Domain alignment = movement in fingerprint / property distributions, not improved activity or solubility.

SETUP

# Fine-tuning Configuration

| Setting | Default | Purpose |
|---|---|---|
| Model | GenMol V2 | extended / bracket SAFE |
| Global batch | 64 | ~14 optimizer steps / epoch |
| Learning rate | 1e-4 | adapt without overwriting pretrained chemistry |
| Warmup | 50 steps | short warmup for small dataset |
| Max training | 25 epochs | cap ~375 optimizer steps |
| Validation | every epoch, 4 samples | stabilize val-loss ranking |
| Early stopping | patience 5 | stop after 5 epochs w/o improvement |
| EMA decay | 0.99 | track short-run adaptation |
| Precision / HW | bf16 / 1 A100 | target execution environment |

## Selection logic

Lowest val-loss EMA weights saved as best.ckpt and used for generation.

Baseline val-loss recorded before fitting as a step-zero reference.

Regularization option: --freeze-layers N adapts only upper layers + MLM head.

Upstream loader-bug workaround: scalar-indexed dataset, vendored GenMol unchanged.

EXPERIMENT 1 · FULL FINE-TUNING, LR 1E-4

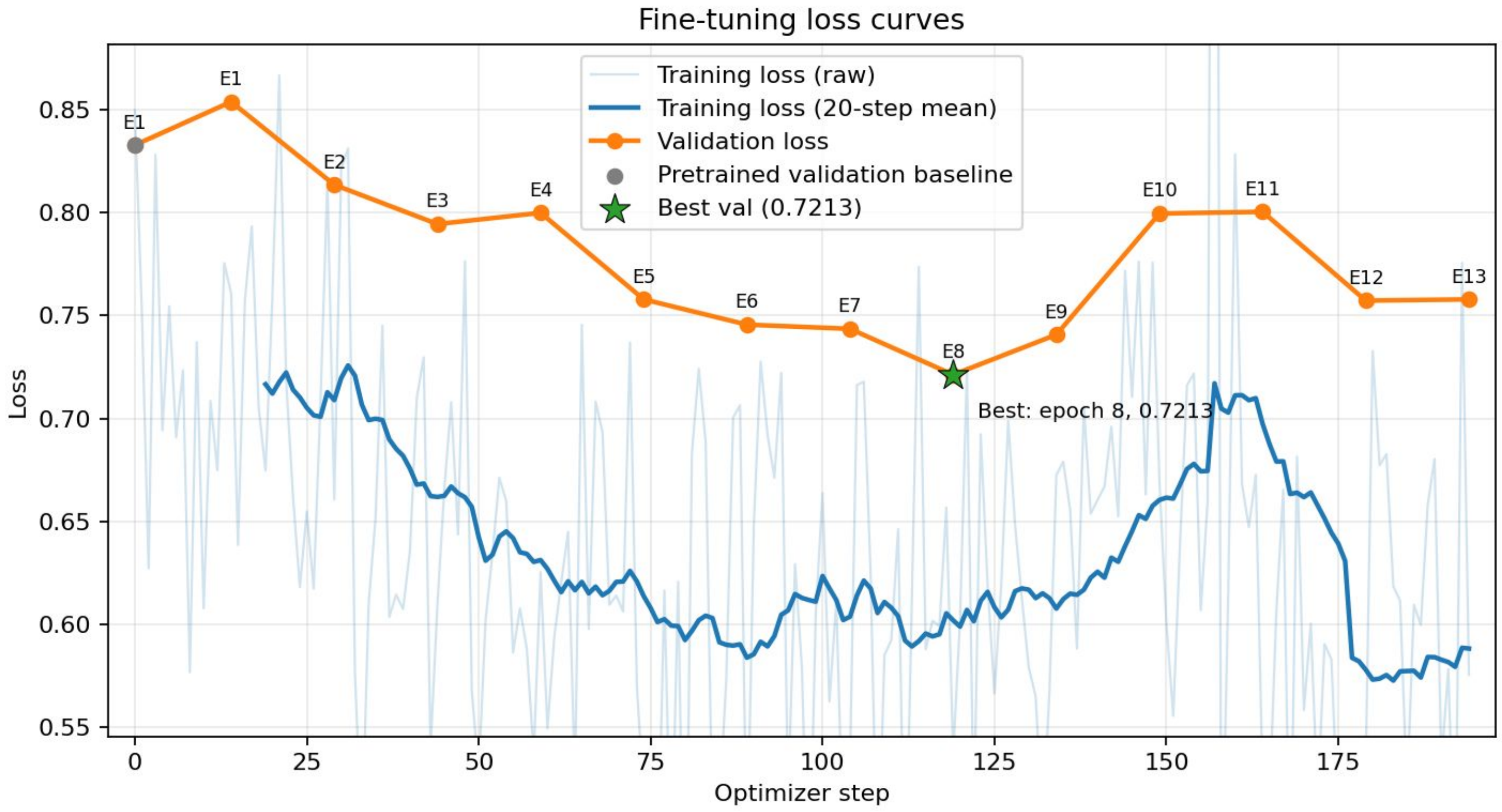
# Base vs Fine-tuned — Quantitative Comparison

| Metric | Base | Fine-tuned | Signal |
|---|---|---|---|
| Validity | 99.3% | 97.5% | 🟢 Preserved |
| Novelty | 93.9% | 85.0% | 🟢 Strong novelty retained |
| Uniqueness | 84.7% | 69.0% | 🟡 Main trade-off |
| Diversity | 0.909 | 0.865 | 🟡 Slightly reduced |
| QED | 0.572 | 0.664 | 🟡 Simplification, not alignment |
| SA score | 2.394 | 1.765 | 🟡 Simplification, not alignment |
| Drug-like fraction | 34.1% | 41.7% | 🟡 Confounded w/ simplification |
| Nearest-validation similarity | 0.203 | 0.289 | 🟢 Clear domain shift |
| FCD-to-validation | 24.44 | 22.40 | 🟢 Improved domain alignment |

**Best val loss 0.7213 at epoch 8 (baseline ≈ 0.83).** Similarity (+0.086) and FCD (−2.0) are the genuine alignment signals; QED / SA gains reflect simplification.

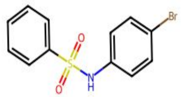
EXPERIMENT 1

# Training & Validation Loss — Full Fine-tuning

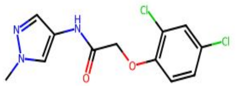



- Validation loss falls substantially, reaching its minimum at epoch 8, then degrades — classic small-data overfitting.
- Early stopping (patience 5) + best.ckpt selection capture the min-val-loss snapshot.

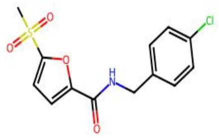
# Five Example Generated Molecules



QED 0.95 | SA 1.48 | sim 0.28

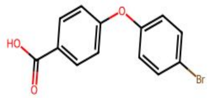


QED 0.94 | SA 1.94 | sim 0.24



QED 0.94 | SA 2.07 | sim 0.25

QED 0.94 | SA 1.34 | sim 0.25



QED 0.93 | SA 1.47 | sim 0.28

| SMILES | QED | SA | Sim |
|---|---|---|---|
| O=S(=O)(Nc1ccc(Br)cc1)c1ccccc1 | 0.945 | 1.48 | 0.276 |
| Cn1cc(NC(=O)COc2ccc(Cl)cc2Cl)cn1 | 0.944 | 1.94 | 0.242 |
| CS(=O)(=O)c1ccc(C(=O)NCc2ccc(Cl)cc2)o1 | 0.939 | 2.07 | 0.254 |
| COc1ccc(NC(=O)c2ccc(Br)cc2)cc1 | 0.939 | 1.34 | 0.250 |
| O=C(O)c1ccc(Oc2ccc(Br)cc2)cc1 | 0.932 | 1.47 | 0.276 |

Sim = nearest-validation ECFP4 Tanimoto similarity. All five are novel, valid, and drug-like.

# Experiment Comparison — LR & Layer Freezing

| Metric (direction) | Original LR 1e-4 | Lower LR 2e-5 | Frozen-9 layers |
|---|---|---|---|
| Validity ↑ | -1.8 pp | +0.7 pp | -1.8 pp |
| Uniqueness ↑ | -15.7 pp | -12.6 pp | -19.2 pp |
| Novelty ↑ | -8.9 pp | -6.6 pp | -12.4 pp |
| QED ↑ | +0.092 | +0.055 | +0.055 |
| SA ↓ | -0.629 | -0.316 | -0.647 |
| Diversity ↑ | -0.044 | -0.016 | -0.034 |
| Drug-like fraction ↑ | +7.6 pp | +1.4 pp | +1.5 pp |
| Nearest-validation similarity ↑ | +0.086 | +0.044 | +0.065 |
| FCD-to-validation ↓ | -2.042 | -2.232 | -1.478 |

**Original LR: strongest property / domain shift.  Lower LR: best diversity + FCD.  Frozen-9: strong SA but largest novelty/uniqueness loss.**

**Recommended: full fine-tuning at original or lower LR; freezing not preferred.**

READ RESULTS WITH CARE

# Limitations

- Small, broad dataset (1,056 mols / 105 val) — variance and overfitting / mode-collapse risk
- Weak domain definition — ESOL is not a therapeutic series; solubility labels unused
- Limited generalizability — specific to GenMol, ESOL, this scaffold split
- Externally controlled length — isolates tokens but doesn't test length learning
- Proxy-only chemistry — QED/SA/FCD ≠ potency, ADMET, synthesizability
- Sampling sensitivity — temperature / randomness / min length shift metrics
- Stochastic uncertainty — bf16/CUDA limit exact repeatability; needs multi-seed
- Validation set double-used — drives both selection and alignment metrics (optimistic bias)
- No uncertainty quantification — single-seed point estimates, no confidence intervals
- Small FCD reference — 105 molecules is well below typical FCD sample sizes

WHERE TO GO FROM HERE

# Recommended Next Steps

| # | Action | Decision / Output |
|---|---|---|
| 1 | Replicate across ≥3 seeds; fixed diffusion-time grid | Verify stability of ranking and gains |
| 2 | Expand freeze-layers × lr Slurm sweep | Select via health gate + composite score |
| 3 | Sweep softmax_temp × randomness (both length priors) | Separate fine-tuning gains from sampling |
| 4 | Replace broad ESOL with a sharper domain | Soluble subset / congeneric / ChEMBL target |
| 5 | Compare full vs freeze vs LoRA/PEFT | Same seeds + evaluation protocol |
| 6 | Add chemistry checks + generation guardrails | PAINS / Lipinski / retrosynthesis / ADMET |
| 7 | Split off an untouched test set | Report final similarity + FCD on test |
| 8 | Bootstrap confidence intervals on all metrics | Attach CIs before claiming improvement |

# Conclusion

- GenMol can be fine-tuned on ESOL on one A100 via Slurm, measurably changing the generated distribution.
- Full fine-tuning at 1e-4 gave the strongest QED / SA / drug-like / similarity gains while retaining 97.5% validity and 85.0% novelty.
- Adaptation traded off uniqueness and diversity; nearest-validation similarity (+0.086) and FCD (−2.0) are the genuine alignment signals.
- Lower LR preserved diversity and achieved the best FCD; freezing 9 layers did not improve the trade-off.
- Feasibility demonstrated — multi-seed evaluation and an independent test set are required before stronger claims.

**PoC successful — approach validated end-to-end.**